

DESTINATION EARTH

HEALPIX AND CLIMATE DT DATA PROCESSING

Climate DT – Finnish user workshop 2026



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Destination Earth

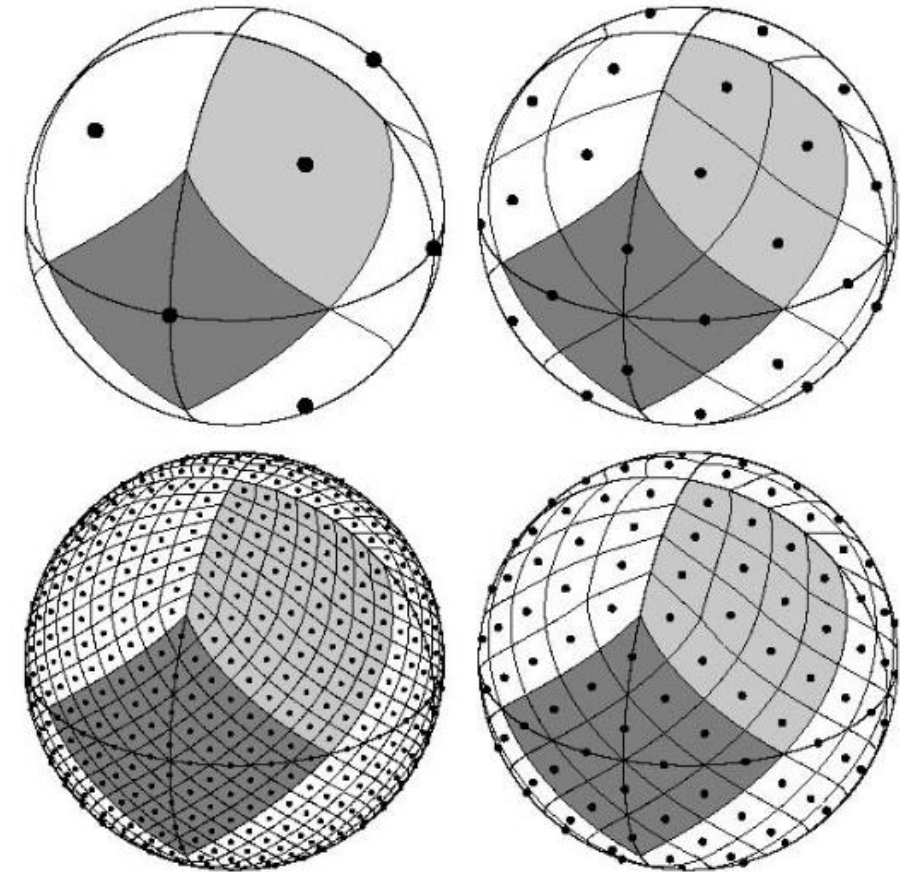
implemented by





HEALPix

- The Climate DT data is interpolated onto a HEALPix grid (the file is still stored as grib file)
- HEALPix: **H**ierarchical **E**qual **A**rea iso**L**atitude **P**ixelation of a sphere
- As name says → Pixelization of data onto a sphere
- Originally created for astrophysics

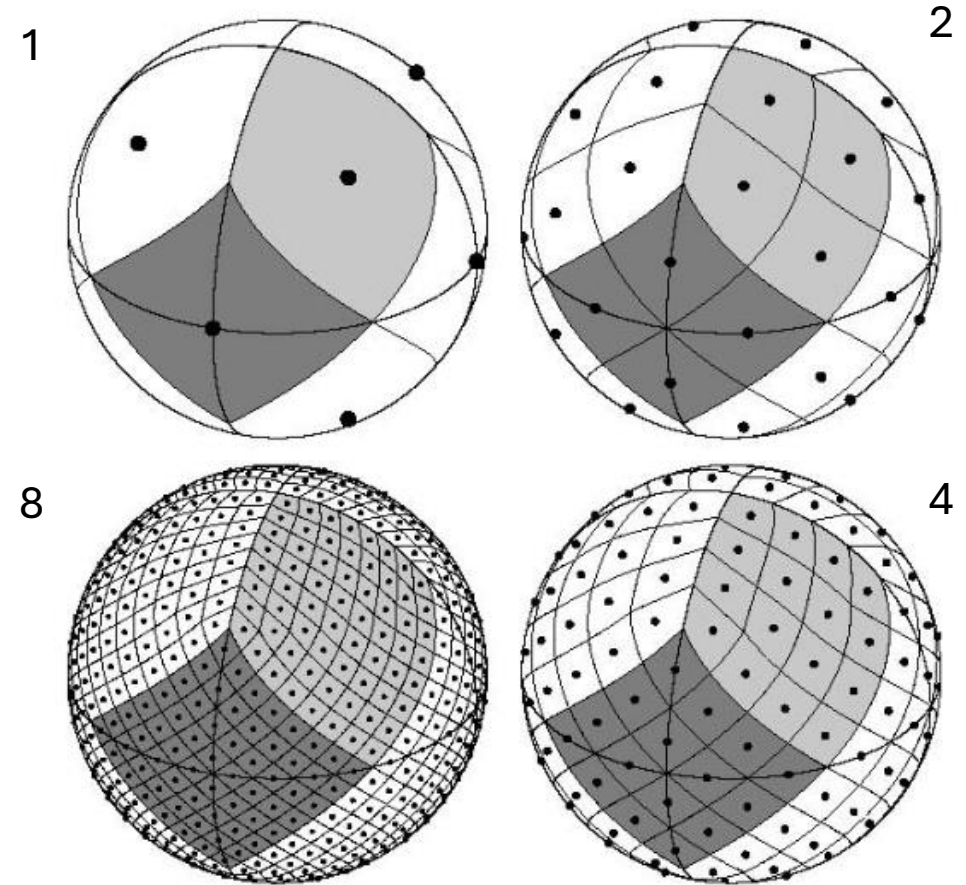


HEALPix partition of the sphere showing different resolutions (N_{sides}). (Górski et al. 2005)



HEALPix

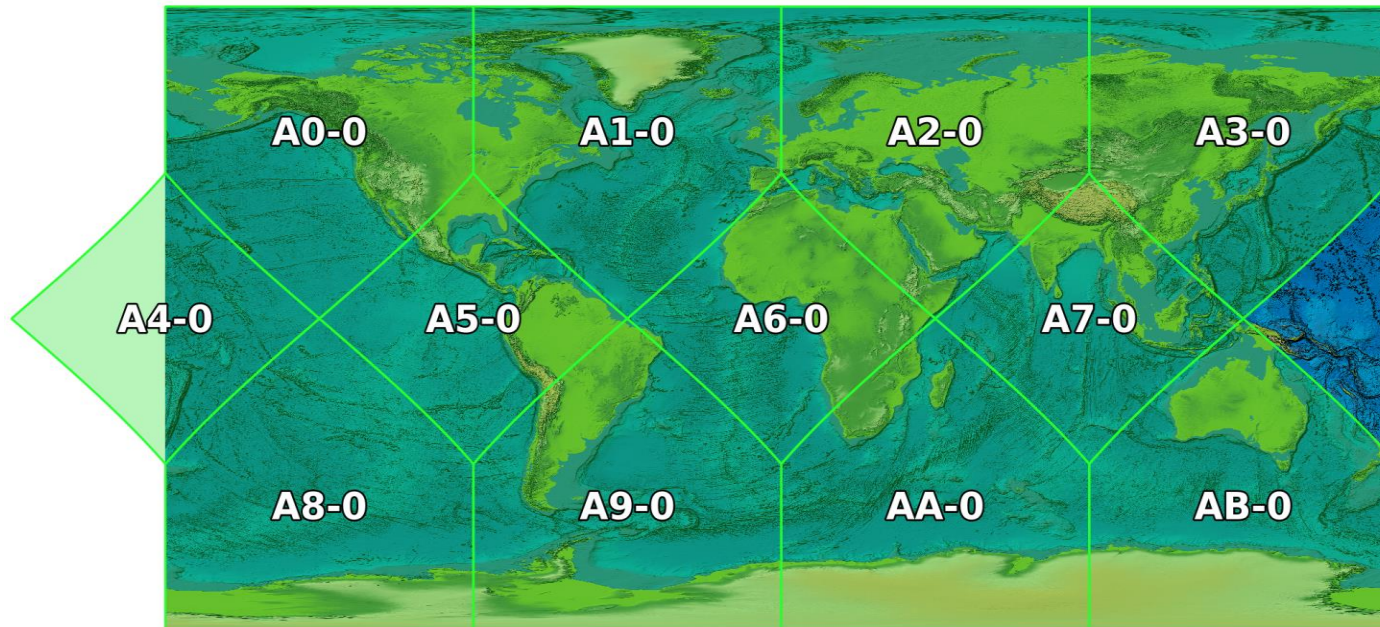
- Lowest resolution has 12 base pixels ($N_{\text{side}}=1$)
- For a resolution parameter N_{side} , the pixels are laid out on $4N_{\text{side}} - 1$ isolatitude rings
- Increase of resolution by division of each pixel by four new ones
- Areas of all pixels at a given resolution are identical
- Pixels distributed along lines of constant latitude



HEALPix partition of the sphere showing different resolutions (N_{sides}). (Górski et al. 2005)



HEALPix levels



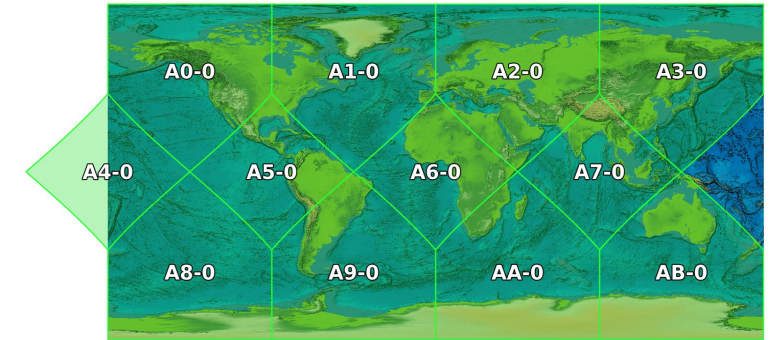
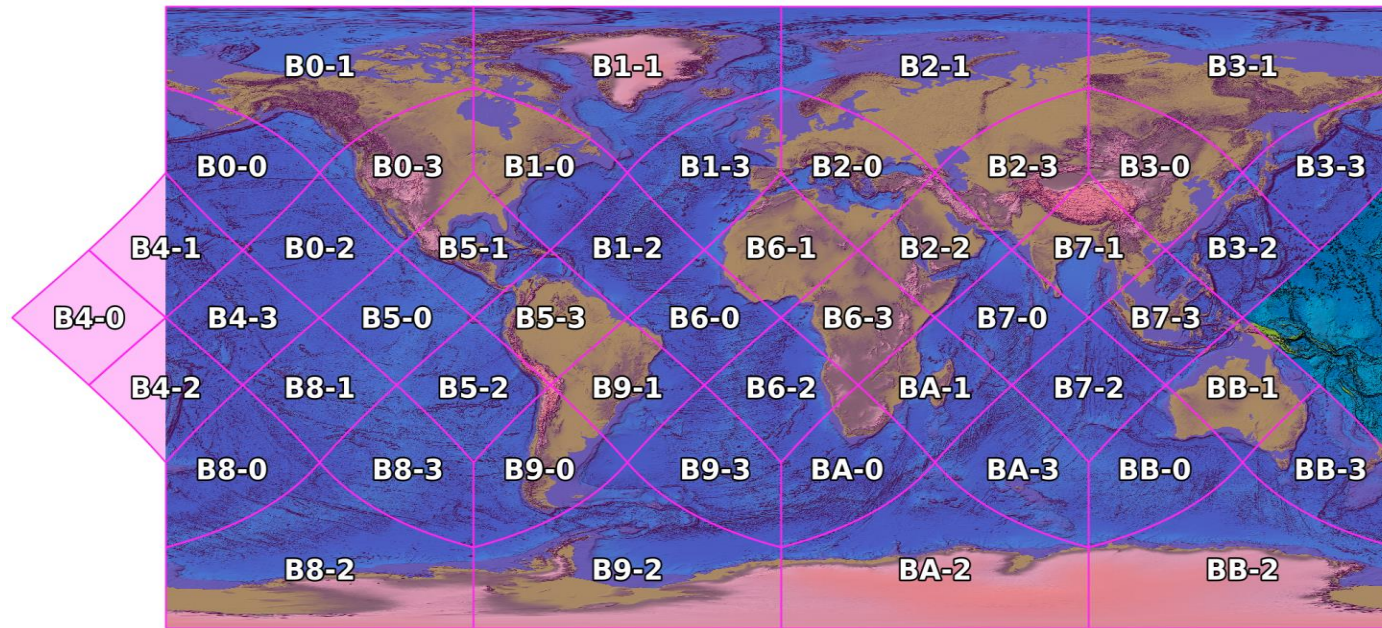
Different numbering schemes can be used to identify pixels (here only one shown)

How do these different resolutions look on Earth?

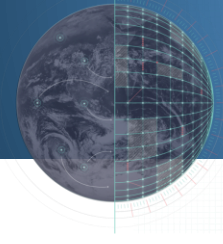
The HEALPix global grid at level 0 (12 shown on a Plate carrée projection)



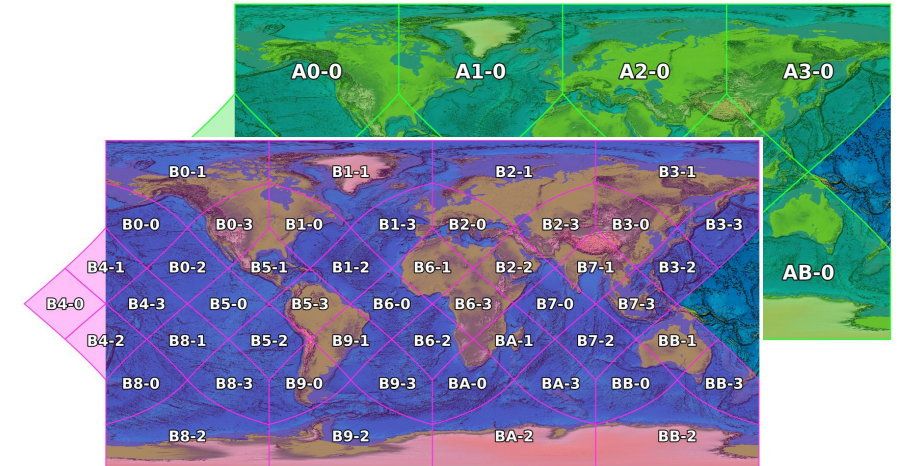
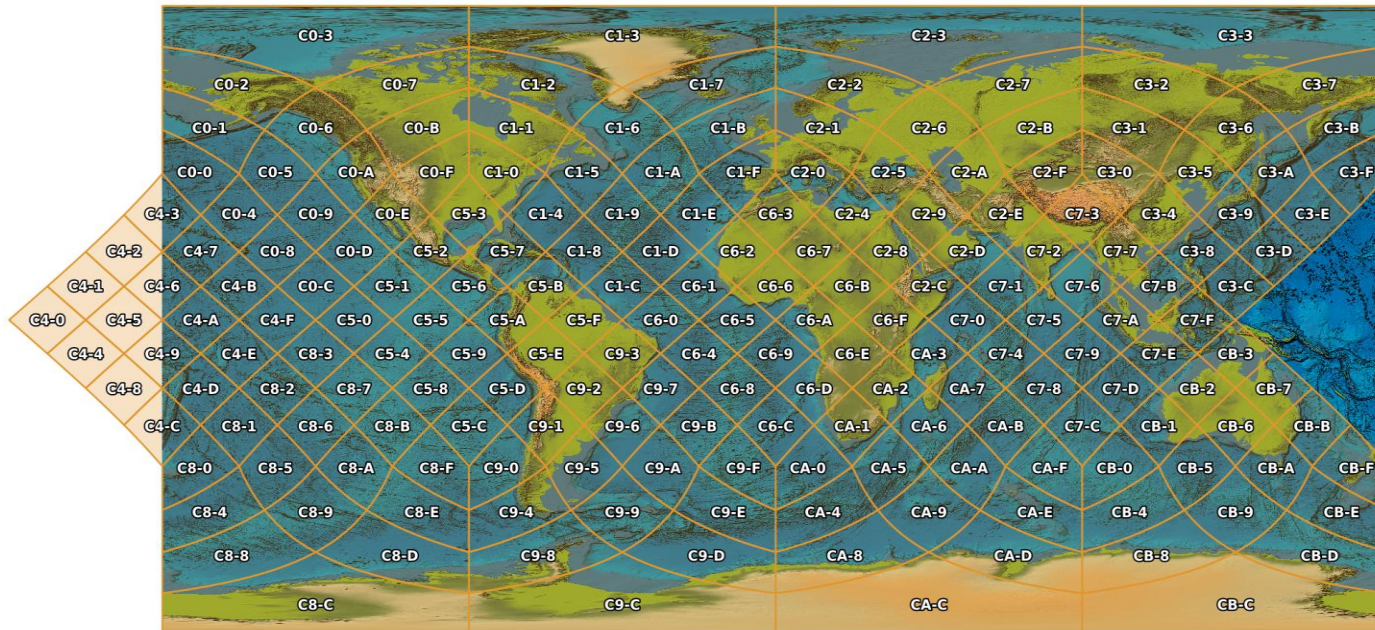
HEALPix levels



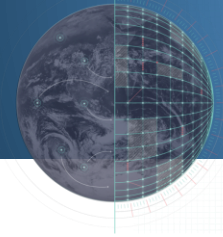
The HEALPix global grid with level 1 zone identifiers shown on a Plate carrée projection



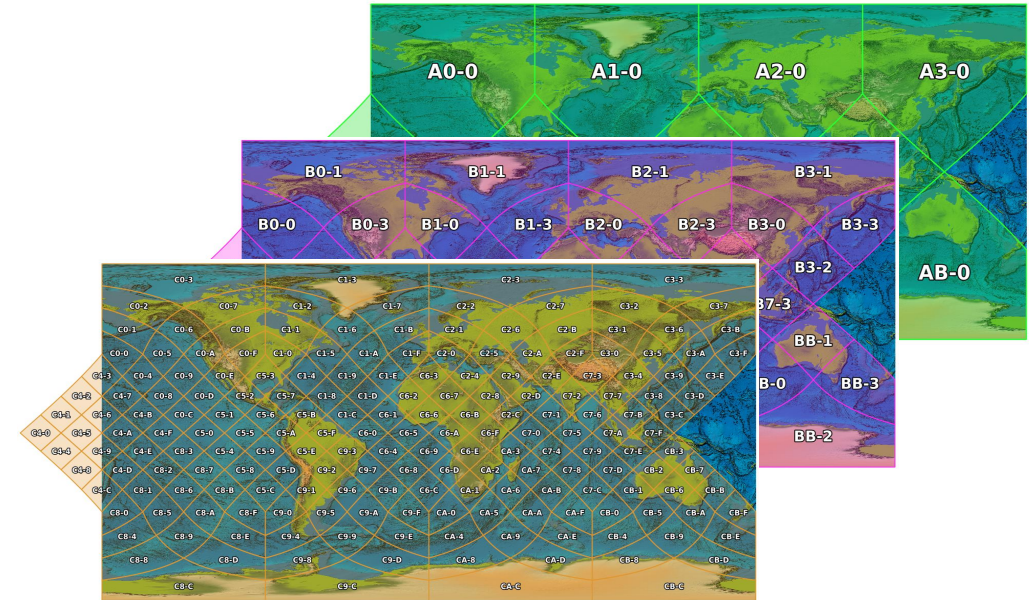
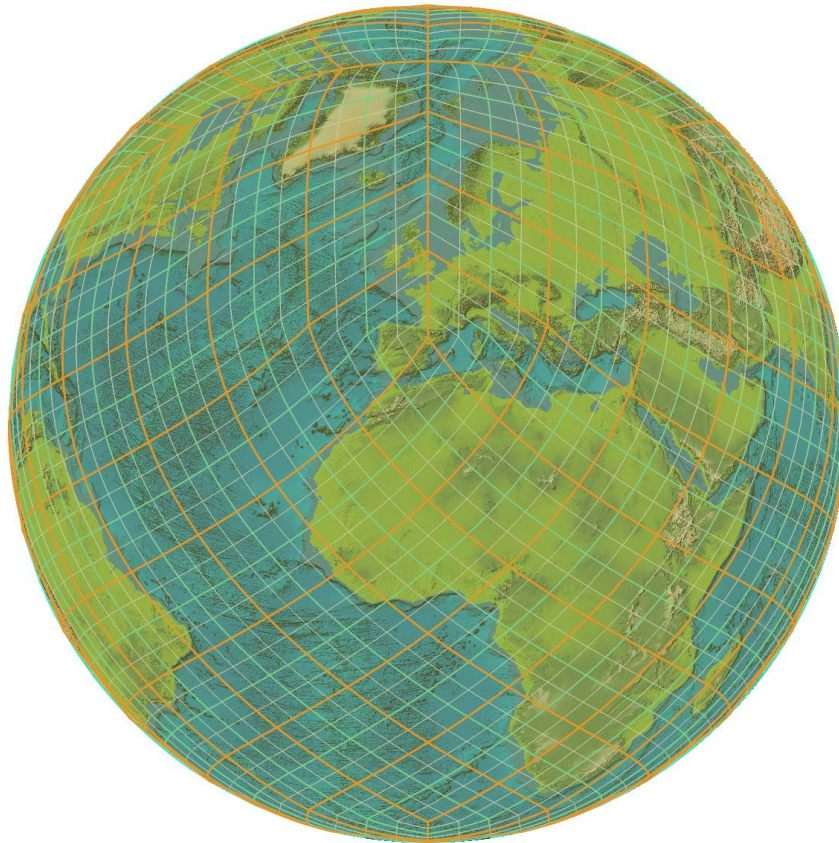
HEALPix levels



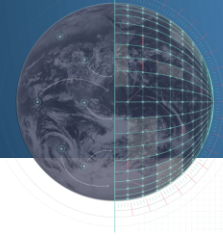
The HEALPix global grid with level 2 zone identifiers shown on a Plate carrée projection



HEALPix levels



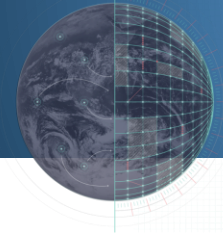
The HEALPix global grid levels 2, 3 and 4
shown on a globe



Climate DT HEALPix grids

Model resolution	FDB encoding	HEALPix zoom level	Nside value (resolution)	Angular resolution	Resolution on Earth in km
5km	high	10	1024	~3.4 arcmin (0.057°)	~6.4 km
10km	high	9	512	~6.9 arcmin (0.11°)	~12.7 km
5 and 10km	standard	7	128	~27.5 arcmin (0.46°)	~51 km

Note: In Climate DT a **NESTED** numbering schemes is used. This is relevant when visualizing data.



Working with HEALPix

Some Python packages for this session:

- Xarray
- Healpy → <https://healpy.readthedocs.io/en/latest/index.html#>
- Earthkit-geo, earthkit-data, earthkit-plots, earthkit-regrid (<https://earthkit.ecmwf.int/>)
- Cartopy

Where to find more about HEALPix: <https://github.com/pangeo-data/awesome-HEALPix>

Górski et al. HEALPix: A Framework for High-Resolution Discretization and Fast Analysis of Data Distributed on the Sphere. The Astrophysical Journal. 2005.



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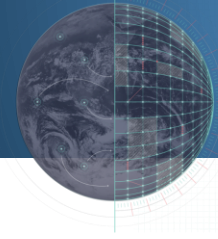
ECMWF



esa



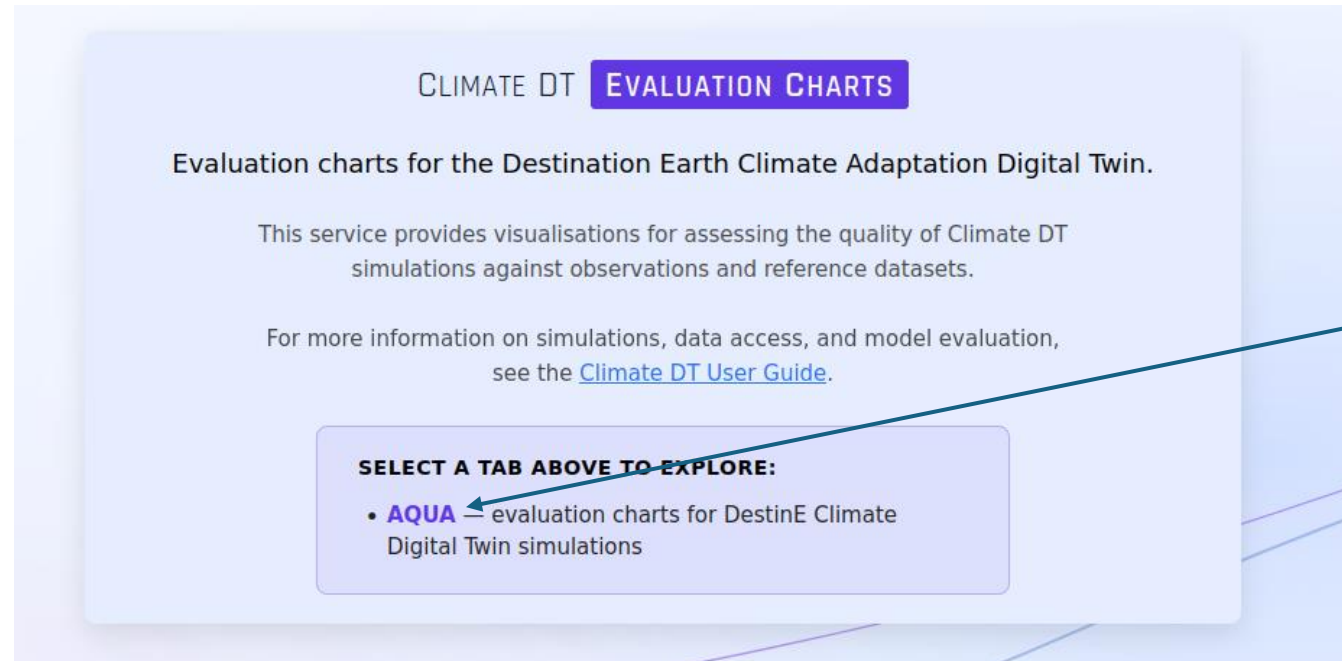
EUMETSAT



Climate DT Evaluation Charts

Exploring the Climate DT Evaluation Charts!

Go to: <https://climatedt-evaluation-charts.destine.eu/>



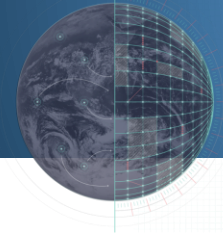
Click on "AQUA"



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Climate DT Evaluation Charts

AQUA

climatedt-gen2



AQUA DESP Diagnostics Explorer

This page summarizes results from AQUA diagnostics for DestinE Climate Digital Twin simulations. Choose an experiment from the left experiments side menu.

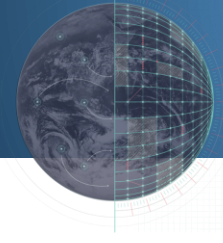
Click on dropdown menu
"climatedt-gen2"



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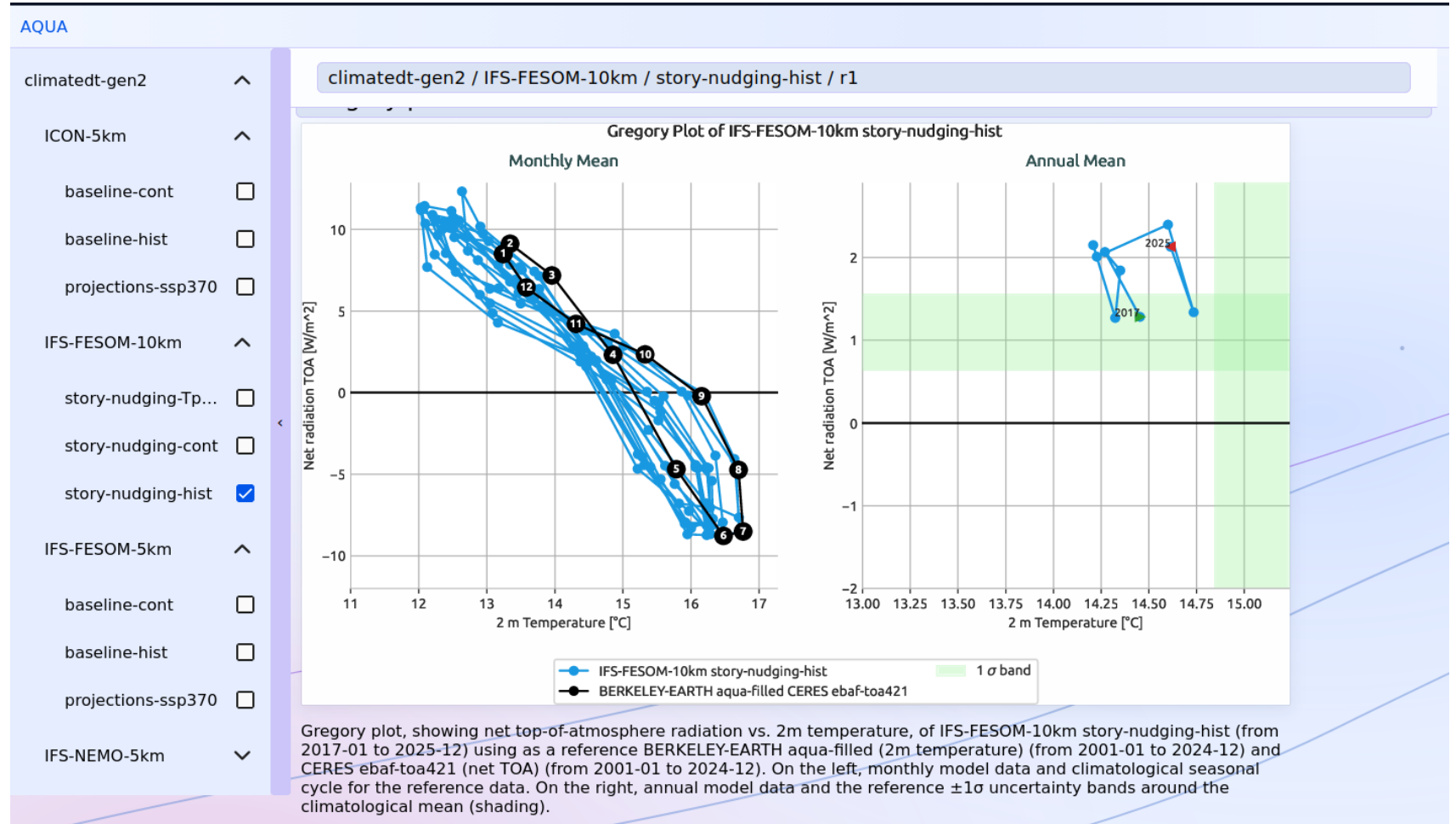
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Climate DT Evaluation Charts

You can click through the different experiments and look at existing figures!

Note: The **experiment IDs** in the Climate DT user-guide correspond to the naming scheme used in the Climate DT Evaluation Charts

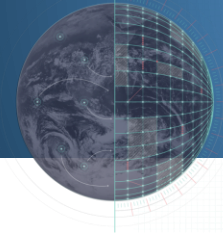




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Climate DT Evaluation Charts

Task 1: Find the North Atlantic Oscillation (NAO) correlation plot for the IFS-FESOM historical 5km plot.

Task 2: Find the time series of the sea surface temperature for the ICON control at 5km.

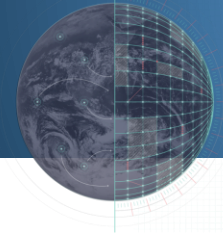
Task 3: Find the annual total precipitation rate bias plot for the IFS-NEMO SSP3-7.0 projection at 5km.



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Housekeeping

Agenda (Times are approximate)

- 09:30 – 10:00 Welcome and Climate DT intro
- 10:00 - 10:05 Climate DT infrastructure and data introduction
- 10:05 – 10:30 Data access demos and hands-on work
- 10:30 – 10:40 Coffee break
- 10:40 – 11:30 Data access demos and hands-on work continued

- 11:30 – 12:30 Lunch break in Plate (reservation exists)
- 12:30 – 12:40 Theory HEALPix grids
- 12:40 – 14:00 Hands-on session 2
- 14:00 – 14:30 Pulla and coffee
- 14:30 – 16:30 Hands-on session 2 continued

Your hosts today: Outi Sievi-Korte, Juha Tonttila, Devaraju Narayanappa, Tracy Kiszler and Antti Vartiainen